

Rectifier MR483000SG3-EF

MEGMEET

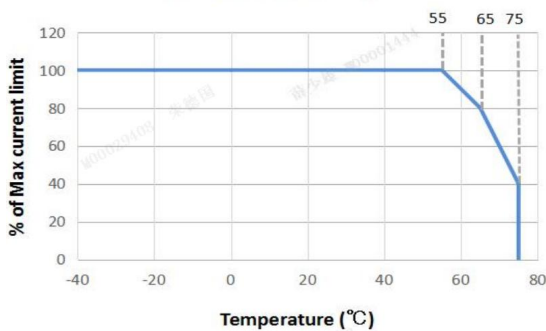
MR483000SG3-EF is a high efficiency, high power density digital rectifier module .It can convert 85- 290V AC input to the default output of 53.5V DC. It has the performance of high reliability, high conversion efficiency, low noise, soft start, hot swap.



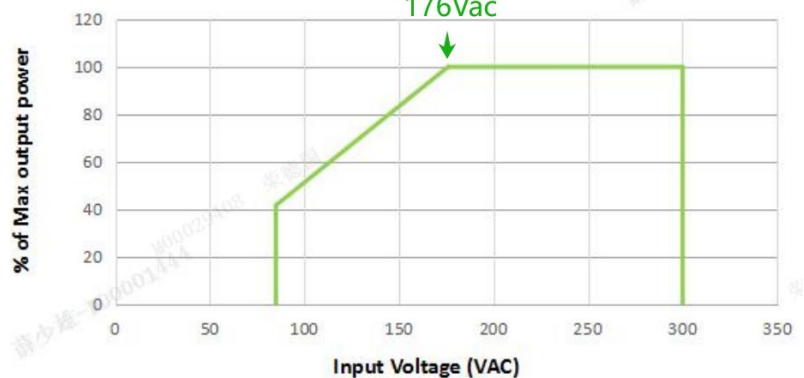
Key Fetures

AC input		DC output	
Input voltage (nominal)	220 Vac / 230 Vac	Output voltage	-42~-58 Vdc, rated: -53.5 Vdc
Input voltage range	85~300 Vac	Peak efficiency	> 97%
Max input current	18 A	Maximum capacity	3000 W
Input frequency	45~65 Hz	Maximum output current	62.5A
Power factor	≥0.99 (220 Vac @20%-100%load)	THD	≤ 3% @ 50%~100% load; <8%@20%load
Control and monitoring		Environmental	
Communication protocol	CAN	EMC	EN 55032 Class B; EN 55035; IEC 61000-3-2; IEC 61000-3-3
LED	Green - Normal; Yellow - Alarm; Red - Fault	Safety	CE, UL, UL-CB UL 62368-1; IEC 62368-1,EN62368-1

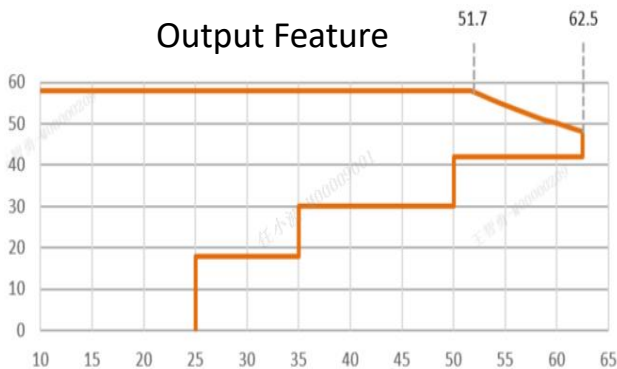
Current limit vs Temperature
176Vac≤Vin≤300Vac



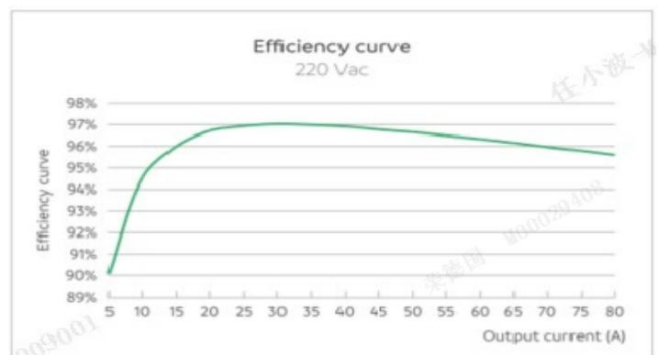
Output Power vs Input Voltage
-40°C≤Temp≤55°C



Output Feature



Efficiency curve
220 Vac



Parameters

Item		Description
Features • Hot swap • Supports intelligent electric meter • Supports CAN communication • Supports LED display • Supports voltage adjustment,current adjustment,and current sharing • Meet Rohs requirement		
Mechanica	Dimension	291mm(L)*106.4mm (W) *41.5mm (H)
	Weight	<2Kg
	Cooling mode	Forced air cooling, built-in fan (with linear speed regulation function)
Input Protection	Over voltage	290Vac<Input voltage<300Vac: alarm light on, output on 300Vac<Input voltage: output off
	Over-voltage recovery	Input voltage<300Vac, automatic recovery
	Under-voltage	Input voltage<80±5Vac output off
	Under-voltage recovery	Input voltage:80~90VAC, automatic recovery
Output Protection	Output over-voltage	Internal over-voltage:58.5VDC~60.5 VDC, Can be set via monitor The module locks out when the internal over-voltage protection occurs
		External over-voltage: >63VDC, And last for more than 500ms The module locks out when the external over-voltage protection occurs
		The above two output over-voltages can be recovered by repowering
	Current limiting	Refer to output characteristic curve
	Short-circuit	Long time short-circuited tolerance, can be automatically recovered after the short-circuit disappears.
	Over temperature	With over temperature protection
Surge Protection	Lightning protection	5KA 8/20μs
Operating environment	Operating temperature	-10°C~+75°C Note: At -10°C ~ 55°C, it works normally; At 55°C ~ 65°C, the output power is linearly derated to 80%; At 65°C ~ 75°C, the output power is linearly derated to 40%,. At -40°C, can power on normally after the input powered on. Output voltage waveform is not required.
	Storage temperature	-40°C~+75°C
	Relative humidity	5%~95% (no condensation)
	Altitude	≤4000m (from 3000 m to 4000 m, the operating temperature decreases by 1°C for each additional 200 m)
Reliability	MTBF	>500,000 hours (25°C)
Audible Noise	Noise	≤55dB(A) (40°C)
Voltage does not destroy the rectifier		0-415VAC